Auditing the Winner: Critic Disagreement Adds No Detectable Failure Signal Beyond the Selector on SWE-Agent Trajectories

Working draft v0.9 — SilentFail-Bench; two benchmarks, 46 repositories analysed, pre-registered throughout

Joseph Morante
Algoverse AI Research

September 24, 2026

Abstract

When an agent commits the best of n attempts under a critic, the committed trajectory is where the critic's confidence and its errors coincide. A common proposal is to audit it with a disjoint panel of critics and escalate on disagreement. We pre-registered and ran this test with executed gold on SWE-bench Verified (41 instances, 9 repositories) and SWE-bench Multilingual (114 instances, 37 repositories): best-of-4 selection by a reward model, a checkpoint-disjoint panel, and the official test harness. Silent failures — committed, agent-declared-complete trajectories that fail hidden tests — were 35% and 45% of the population. On the registered endpoint — the out-of-fold gain in AUROC from adding panel disagreement to the selector's own score — disagreement adds nothing detectable: pooled over 47 failures and 46 repositories, the three estimators that use every row give -0.03, -0.01 and $+0.01$, every interval includes zero, and the loosest pooled upper bound is $+0.12$. Disagreement alone ranks failures at AUROC 0.51 and 0.57, indistinguishable from the selecting critic's own score, and this held across 4k–16k contexts, pruned overlapping windows, eight reward models from 1B to 32B including two code-trained ones, two re-selection regimes, and a single-family panel of LLM judges. Point estimates that cleared a structural baseline on Verified (windowed disagreement 0.69; patch size 0.67; one 1B critic 0.71) all fell to chance on 37 repositories: no configuration produced a replicable signal. An unselected-candidate control shows that with this selector the committed candidate fails no more often than one passed over, while the selector's own score ranks failures at 0.81 among unselected candidates and 0.61 among committed ones on Verified — the selection-conditional gap measured directly. Re-selecting with `gpt-5`, which ranks failures across the candidate pool at 0.71 on both benchmarks and re-commits three-quarters of the population, leaves the result unchanged: disagreement adds $+0.03$ $[-0.08, +0.14]$. Test–retest shows the critics are stable ($r \geq 0.98$) and content-driven: the failure is in what disagreement measures. We argue that audit signals must be evaluated on the population the selector commits, against the selector's own score, with executed gold; we report an upper bound on the residual value of disagreement; and we document the instrument failures that would otherwise have produced a positive result.

In plain terms. A coding agent tries a task several times and a scoring model picks its best attempt. We checked those picks against the project's real test suite. A third to a half of them were broken — the agent said it was finished, the scorer agreed, and the tests failed. The idea we tested was a safety net: ask several other scoring models to grade the same attempt, and treat *their disagreement* as a warning that something is wrong. It does not work. Disagreement flags broken work no better than a coin flip, and adds nothing to what the original scorer already told you. The scoring models themselves are not the problem — they are consistent, and the best of them spots roughly two-thirds of the failures on its own. The problem is that measuring how much they *differ* throws that information away. We also show why this is easy to get wrong: several promising-looking results in our own pilot disappeared when we re-ran everything on three times as many code repositories, and six bugs in our own measuring equipment would each have produced a convincing but false finding.

1 Introduction

Best-of-n selection under a learned critic is now a default in agent scaffolds: generate several trajectories, score each with a reward model, commit the argmax. Three claims about this are easy to conflate, and this paper keeps them apart. The *theory* is that argmax selection on a noisy score induces the optimizer's curse [46]: among candidates, the top-scored one is disproportionately the one whose flaws the critic cannot see, so the score carries less information about the committed candidate than about the pool. What we *observe* is one instance of that theory measured directly: on SWE-bench Verified the selecting critic ranks failures at 0.81 among the candidates it passed over and at 0.61 among the ones it committed, on the same instances. What we do *not* show is that selection raised the failure rate of the committed candidate — with the selector we used, it did not (§4.1). A committed trajectory that fails its task while carrying a high score is a *silent failure*; they are a third to a half of what this agent committed, whether or not selection put them there.

The obvious safeguard is a second opinion — or several. If a panel of critics, disjoint from the selecting critic, disagrees sharply about a committed trajectory, that disagreement might carry information the selecting critic has exhausted. The proposal behind this work stated the hypothesis as a falsifiable, selection-conditional estimand and committed in advance to publishing the null.

This paper reports that result. The idea that correlated critics share blind spots is not ours: the correlated-judge literature predicts it [2, 13, 15, 22, 23], and best-of-n reward hacking is well characterized [11, 21, 61]. What had not been done is the measurement: put a disjoint audit panel on the population a critic actually committed, with executed gold, structural baselines and pre-registered bars, and ask what it adds over the selector. Our contributions are:

- **A selection-conditional evaluation protocol for audit signals** — selected population, selector baseline, executed gold, pre-registered bars — with its first two-benchmark instantiation on SWE-agent trajectories. Candidate-pool discrimination does not measure residual auditing value on the selected population [9, 31]; we evaluate on the selected population directly.
- **A replicated null with an estimator-robust upper bound.** On the proposal's registered endpoint — out-of-fold AUROC of a model with the selector's score and panel disagreement over one with the score alone — disagreement's incremental value is indistinguishable from zero under four estimators of it, with a loosest pooled upper bound of +0.12 over 47 positives and 46 repositories, and +0.25 on the smaller benchmark alone (§4.7). The primary protocol's own upper interval is +0.22; across the whole configuration sweep the widest reaches +0.45; positives that appeared on twelve Verified failures did not travel.
- **Diagnostics and an instrument record:** test–retest stability of long-window reward-model scores; the repository-clustering ceiling on effective n for SWE-bench Verified; and six instrument

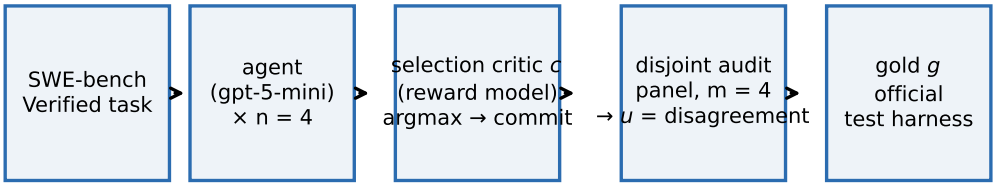


Figure 1: **How the experiment works.** An agent writes several candidate fixes; a scoring model picks one; a separate panel grades the pick; the project's real tests decide whether it was right. c is the selecting critic's scalar score of a whole trajectory; best-of-n commits the argmax. A disjoint panel of m reward models produces u, the sample standard deviation of their rank-normalized scores (rank norms fit on training repositories only). Gold g is the official SWE-bench harness verdict on the committed patch.

failures that each produced a plausible false finding.

Throughout, "critic" means an off-the-shelf reward model or LLM judge used without task-specific training. Trained trajectory verifiers [39, 45, 51] are a different class; we return to them in §4.5.

2 Setup and estimand

Unit and gold. The unit is one committed trajectory τ. Gold $g(\tau) \in \{0, 1\}$ is trajectory-level task success under the official `swebench` 5.x harness; failed tests do not imply every step was wrong, and we make no step-level claims. Positive class throughout is failure (g=0).

Selection and audit. c_{sel} scores the full committed trajectory text; best-of-n ranks n full candidates by this score and commits the top-1. The audit panel $\{c_1, \ldots, c_m\}$ satisfies $c_{\text{sel}} \notin \{c_i\}$ at checkpoint level; u is their disagreement.

Primary population. Committed trajectories whose agent exited with a submission and a non-empty patch — the agent believed it was done. Within it, silent failures are g=0.

Metrics. M1: AUROC(u) for silent failure. M2: AUROC($-c$). M3 = M1 − M2 with a repository-clustered bootstrap (4,000 resamples, groups resampled with the fitted scores held fixed). A held-out logistic ΔAUROC was pre-specified as confirmatory only when ≥ 10 positives fall in the test split; with 41 instances it never did, and we report raw-score AUROCs as registered.

Pre-registration. Population, positive class, metrics, predictions and a stopping rule ("fewer than ten positives yields no conclusion in either direction") were committed to the repository before any gold

existed (`PREREGISTRATION_SWEBENCH_PILOT.md`). Each later analysis was added as a dated amendment before its data existed. Every deviation is declared in §6.

3 Instrument, and what it took to trust it

The pilot runs on a laptop and free GPU tiers: mini-swe-agent 2.4.6 driving `gpt-5-mini` through the OpenAI API ($0.02 per trajectory with prompt caching), SWE-bench images under Rosetta in a Colima VM on Apple Silicon, reward models on Metal or on Kaggle's free T4s, and the official harness for gold.

We record six instrument failures because each would have produced a publishable-looking result, and each was caught by an oracle control or a regression test rather than by inspection of the numbers.

1. A sandbox flag hid `numpy` from the gold harness: every trajectory scored as a failure. Caught because the benchmark's reference solutions are run through the harness before any model patch.
2. An LLM-as-judge selecting critic emitted three distinct values over 640 candidates, so best-of-16 silently became first-draw: 39 of 40 committed trajectories were identical to the n=4 run. Selection had not happened in the earlier pilots. Caught by the chosen-index histogram.
3. On a fresh VM the harness's image pulls failed on an architecture manifest: 41/41 failures in 499 seconds. Caught by the harness's own error count.
4. Our power function reported ≈0.40 "power" under the true null. Caught by simulating the null; replaced by a null-calibrated clustered bootstrap.
5. Repository grouping truncated hyphenated repository names, corrupting the held-out split. Caught by a regression test on the identifier parser.
6. Rank normalization was fit on all rows, leaking test rows into the panel statistic. Caught by a test that fits on the training split only. Every u reported in this paper is post-fix: recomputing the stored values from raw panel scores with training-repository normalization reproduces them exactly (max absolute difference 0.0 on both benchmarks), while the leaky computation differs by up to 0.023 and would have given AUROC(u) = 0.523 rather than 0.508 on Verified.

The rule we adopted: a rate of exactly 0% or 100%, or a statistic that cannot move, is an instrument, not a result. The replication added two engineering stalls that produced no false finding (a per-image build layer and an API rate limit), recorded in the repository.

4 Results

4.1 Silent failures are common among committed trajectories

Fifty instances were sampled with a fixed seed; generation completed for 41 (the deviation is explained in §6). Of 164 candidates, 138 exited having submitted (15.9% hit the 40-step limit); 17.7% produced no patch at all and are excluded. Selection by `Skywork-Reward-V2-Qwen3-4B` committed 34 usable trajectories; the harness resolved 22 and failed 12, with zero errors.

35.3% [21.5, 52.1] of committed, agent-declared-complete trajectories were silent failures on Verified, and 45.5% on Multilingual (§4.4). On HumanEval+ the same pipeline yielded 8% at any selection pressure; SWE-bench supplies the phenomenon in quantity. Selection was real (Fig. 2, top left), and corr(c, u) = 0.11, so u was near-orthogonal to c — the precondition the design wants.

Configuration	AUROC(u)	AUROC($-c$)	M3	95% CI
Primary: 4k prefix, 4 chat RMs	0.508	0.621	−0.114	[−0.37, +0.22]
robustness, m=3 family-disjoint	0.545	0.621	−0.076	[−0.39, +0.23]
Submission window: diff + last 3k tok	0.458	0.530	−0.072	[−0.35, +0.13]
Patch in view (4k), 4 chat RMs	0.494	0.621	−0.127	[−0.28, +0.11]
+ AceCodeRM-7B as 5th auditor	0.473	0.621	−0.148	[−0.30, +0.10]
8B named critic selecting; disjoint panel	0.455	0.602	−0.148	[−0.49, +0.26]
the same at a true 16k (33 rows)	0.483	0.736	−0.252	[−0.47, +0.28]
Code-trained critic selecting; 5-RM panel	0.542	0.472	+0.069	[−0.21, +0.45]
Full trajectory, true 16k, 4 chat RMs	0.439	0.621	−0.182	[−0.38, +0.17]
LLM judges (gpt-5 family), patch in view	0.528	0.621	−0.093	[−0.30, +0.10]
execution-aware (+ last 6k tok)	0.640	0.621	+0.019	[−0.28, +0.16]
Pruned overlapping windows, $u_{\max}$	**0.686**	0.621	+0.064	[−0.11, +0.33]

Table 1: The pre-registered primary and all amendments. Columns are M1, M2 and M3 = M1 − M2 (§2). Same 34 trajectories and 12 positives, except the re-selection rows: the 8B re-selection has 12, and the 16k 8B and code-trained re-selections have 11 (the 16k row excludes one ungradable patch). M2 is the selecting critic's own score at the row's input where it was re-scored, otherwise the primary c. CIs are repository-clustered bootstraps on the paired difference. Only the last row's u clears the structural bar of 0.665 (§4.3); in the M2 column only the 16k re-selection's own score, 0.736, exceeds it.

Whether selection *concentrated* failure — whether the committed candidate fails more often than one the critic did not pick — is a separate question that a committed-only sample cannot answer; Amendment 4f, pre-registered before grading, answers it: one unselected candidate per instance was graded on both benchmarks (31 and 71 paired rows). **The committed candidate failed no more often than one the critic passed over**: −3 points on Verified [−21, 0] and −4 on Multilingual [−11, +3]. With this selector, best-of-4 did not concentrate failure, and we claim a failure *rate*, not concentration. The control also measures the selection-conditional gap directly. On Verified the selector's own score ranks failures among the candidates it did *not* pick at 0.81 [0.74, 1.00], but among the ones it committed at 0.61 on the same instances: selection consumed the critic's own signal, which is the premise of the design. On Multilingual the critic has no signal in either arm (0.43 unselected, 0.51 committed), so there was nothing for selection to remove. Panel disagreement on the unselected arm is 0.37 and 0.51 — no better off-selection than on (Fig. 3).

4.2 The safeguard does not: the ladder of nulls

Table 1 and Fig. 4 are the core of the paper. The primary analysis — four chat-preference reward models auditing a 4,096-token prefix — gave AUROC(u) = 0.508. Every subsequent amendment was declared before its data existed and asked whether some specific limitation of that primary was hiding the signal:

- *The critics never saw the patch.* Scoring the final diff plus the last 3,000 tokens: u fell to 0.458.
- *The critics were too small, or the wrong ones.* Six chat-preference RMs from 1B to 8B, including the proposal's named critic `Skywork-Reward-V2-Llama-3.1-8B`, each scored with the patch in view: as individual detectors of the 12 known silent failures they scored 0.45–0.64, four of six at or below chance; the panel gave 0.494.
- *Selection was by the wrong critic.* Re-selecting with the named 8B critic changed 20 of 34 committed trajectories; graded, the failure rate was again 35%, and u was 0.455. Repeating the re-selection at

a true 16k changed 23 of 33: u 0.483, while the critic's own score on the set it had itself selected reached 0.736.

- *Chat-preference training is the problem.* `AceCodeRM-7B`, trained on code with execution-feedback preferences, scored **0.405** as a detector — below chance — and adding it to the panel left u at 0.473. One registered prediction attached to this critic, A2-P3, asked whether its prefix and full-length scores correlate above 0.5, as a critic with a stable view of quality should. **It is not evaluable**: the model fell back to a 4k window on 31 of the 34 committed rows even on a 24 GB GPU, so its apparent $r = 0.997$ is a score correlated with itself rather than a prefix-versus-full contrast, and its 32B sibling was run only at 8k/16k and has no prefix pass. We report the prediction as attempted and undecidable, and give the selector's own contrast instead: $r = 0.35$ between its 4k prefix and its true 16k score, and $r = 0.18$ between the prefix and the patch-in-view window. As a selector it changed 24 of 34 trajectories; u on that set was 0.542, its own score 0.472. Its 32B sibling, run in fp16 on an 80 GB GPU (131 of 164 rows at 8k, the rest at 16k), scored 0.557 as a detector and 0.50 by mean window.
- *The context was too short.* Re-scored on a 24 GB GPU at a true 16k tokens (every row at the declared length for the six chat RMs; the code-trained model fell back to 4k on 153/164 rows), five of six chat RMs improved as individual detectors — the 1B Skywork model to 0.712 — with paired 4k→16k gains of +0.04 to +0.12 (the sixth, −0.03) whose intervals all include zero. The selecting critic re-scored at 16k reaches 0.587 on its own; Table 1 keeps M2 at its prefix value of 0.621 because that is the score selection actually used and the basis of M3. The panel's disagreement at 16k was 0.439, the lowest of any row. An 8k point shows the ladder is not monotone: the 8k score sits between the 4k and 16k values for only two of the seven critics that have a score at all three lengths (the 32B has no 4k pass).
- *Reward models are the wrong class.* Three LLM judges (`gpt-5`, `gpt-5-mini`, `gpt-5-nano`; one vendor, hence a single-family panel) rated the probability that the patch resolves the issue. Shown the problem and the diff, `gpt-5` reached 0.720 and `gpt-5-mini` 0.752 — above the structural bar — but their disagreement with `gpt-5-nano` was 0.528. Adding the last 6k tokens of the transcript, including the agent's own test runs, did not help (best judge −0.03; disagreement 0.640). `gpt-5-mini` is also the agent, a self-evaluation confound; `gpt-5` is not, and scored 0.72.

Every single-window prediction failed. The intervals on M3 are wide — twelve positives — and a moderate positive effect is not excluded by any one row; the rows are also re-measurements of the same twelve positives, not independent replications. What they establish jointly is narrower than "no detector": under every variation of input length, size, identity, training signal, critic class and selection regime, *the disagreement of a panel on one window of a committed trajectory* carries no information about its hidden failure that the selecting critic's own score does not.

4.3 Windowed auditing: the one row that clears the bar

Amendment 3, pre-registered before any windowed score existed, made one structural change: collapse tool outputs longer than 24 lines to their first and last 8, then score overlapping 3,500-token windows (25% overlap, problem statement prefixed, final window anchored at the patch) and take the panel's maximum disagreement. All three registered predictions passed: AUROC($u_{\max}$) = 0.686 against a bar of 0.665; a gain of +0.18 over the prefix panel against a bar of +0.10; and the named 8B critic's mean-window score, 0.652, against 0.634.

We report this as the study's strongest positive signal and immediately bound it. A free permutation of labels gives $p = 0.04$; permuting within repository, which removes the contribution of repositories

Quantity on SWE-bench Multilingual 77 rows, 35 positives, 37 repositories	AUROC	95% CI / p
panel u, 4k prefix (primary)	0.565	$[0.43, 0.71]$
paired vs the selecting critic	$+0.005$	$[-0.19, +0.22]$
selecting critic $-c$	0.560	—
$u_{\max}$ over pruned windows	0.541	$[0.40, 0.68]$, $p_{\text{repo}} = 0.53$
u_{mean} / u_{last}	0.541 / 0.552	—
lines added (ruler)	0.508	$[0.35, 0.65]$
best single RM at 16k (Skywork-1.7B)	0.577	seven RMs: 0.36–0.58
named 8B critic, mean window	0.465	—
LLM judges: gpt-5 / mini / nano	0.662 / 0.604 / 0.576	each within its CI of c
LLM-judge disagreement u_{LLM}	0.514	$[-0.22, +0.13]$ vs c
registered endpoint, $M_{c+u} - M_c$ (LORO)	-0.022	$[-0.13, +0.08]$

Table 2: Every pre-registered and secondary quantity on the replication.

that both fail more and disagree more, gives $p = 0.21$. Leaving out any one of the 34 instances moves the estimate between 0.657 and 0.731 — across the bar. The paired gain over the selecting critic is $+0.06$ $[-0.11, +0.33]$. Pruning removed 69% of tokens, so most trajectories fit in one or two windows (median two); "multi-window" here mostly means "the pruned trajectory, plus one overlapping cut". In 43% of multi-window rows the maximum fell on a window other than the last, so the middle of the trajectory contributes something. $u_{\max}$ is also a maximum of noisy statistics and is biased upward under the null. A reader who takes 0.686 as a detector is reading more than twelve positives can support; a reader who dismisses it is ignoring the one configuration whose pre-registered bars it cleared. Other registered predictions also passed — the premise check, the LLM-judge single-detector bar, and the replication's null-replication bar — but this is the only one asserting that *disagreement* detects failure.

4.4 Replication on 37 repositories: the null survives, the exceptions do not

The binding limitation of everything above is nine repositories in the analysed set, drawn from a benchmark of twelve. Amendment 4e, pre-registered before any trajectory existed, re-ran the instrument on SWE-bench Multilingual: 41 repositories in nine languages, up to three instances each (114; 37 repositories survive into the analysed population), k=4 (456 trajectories), the same selection critic and panel, agent sandboxes and gold on Modal (three reference-patch / empty-patch oracle pairs passed before any model patch was graded). Of 114 committed trajectories, 77 formed the population (30 empty patches, 7 ungradable); **35 were silent failures (45.5%)** across 37 repositories — three times the positives of the pilot, with an inference floor of 20 pre-registered.

The three registered predictions resolved as follows. **R-P1** (the null replicates): panel u on the 4k prefix, 0.565 [0.43, 0.71], against the critic's own 0.560; paired difference $+0.005$ $[-0.19, +0.22]$; within-repository permutation $p = 0.85$. *Passed.* **R-P2** (the windowed positive confirms): $u_{\max}$ 0.541 [0.40, 0.68], within-repository $p = 0.53$; leave-one-instance-out range 0.528–0.557 — the interval that on Verified straddled the bar now sits tightly on chance. *Failed.* **R-P3** is vacuous: $u_{\max}$ does exceed the lines-added baseline, because the baseline itself fell to 0.508 [0.35, 0.65]. Patch size predicted silent failure on Verified and does not here; it was a regularity of that benchmark, not of agents.

Table 2 and Fig. 6 carry the point. Nothing that rose above the ruler on Verified rose above chance on Multilingual: the best reward model at 16k is 0.577 (the 1B model that scored 0.712 is at 0.485); the named 8B critic 0.52; the code-trained model 0.356; every windowed single-critic aggregate 0.45–0.64. With 35 positives the intervals are half the width they were, and they now exclude the effects the

pilot's point estimates suggested.

4.5 What the critics do and do not carry

On Verified, single critics given enough of the solution reached 0.71–0.75 (the 1B reward model at 16k; `gpt-5` and `gpt-5-mini` with the patch in view), while every disagreement statistic built on the same scores landed at 0.44–0.64. On Multilingual the reward-model half of that observation did not survive (§4.4), and the LLM-judge half shrank: with the patch in view, `gpt-5` 0.662, `gpt-5-mini` 0.604, `gpt-5-nano` 0.576 — each within its interval of the selecting critic — and their disagreement 0.514. The strongest single detector in the whole study is a frontier judge reading the diff, at two-thirds; nothing built from disagreement is above chance on either benchmark. Whatever the critics carry, disagreement is not extracting it. A later, exploratory check sharpens this: adding two judges from a newer model generation (`gpt-6-luna`, `gpt-6-sol`) raises the best single detector in the entire study to 0.776 [0.68, 0.87] on the replication — both beat every `gpt-5` judge on both benchmarks — while the disagreement of that newer pair falls to 0.41 on both, *below* the weaker trio's 0.51–0.53, and a five-judge panel gives 0.44 on 35 positives. Better critics; no better disagreement. The gap widens rather than closes as the individual judges improve. Those judges also rank considerably better than they know: every one is overconfident, stating resolve probabilities averaging 0.66–0.84 against actual rates of 0.55–0.65, with expected calibration error of 0.16–0.34 and several Brier scores at or worse than the 0.25 obtained by always answering "50%". The best ranker, `gpt-6-luna`, is also the most overconfident. Anyone wanting a probability rather than a ranking would have to recalibrate first.

Two things we did not test bound the claim from above. First, the selector is a 4B chat-preference reward model, not a deployed-grade trajectory verifier; whether a stronger selector leaves a different residual for an auditor is open, and Shum et al. [45] provide the natural instrument. Second, *trained* auditors exist and report better numbers on related populations: a 4B verifier trained on labelled failures selects among candidate trajectories and outperforms frontier judges at finding them [39]; a 0.6B prefix monitor predicts SWE-agent failure at a 5% false-positive rate [51]; lightweight statistical detectors beat LLM judges on agent-claimed false success [1]. Our result is about what a panel of *off-the-shelf* critics contributes through disagreement, evaluated the way an escalation trigger would be used; it says nothing against training an auditor, and the protocol here is how such an auditor should be evaluated. Second, a pre-registered incremental-value test (Amendment 4a) asked whether u or c adds anything to a structural model of the diff and transcript under leave-one-repository-out evaluation: u adds -0.03 $[-0.08, +0.04]$; c adds $+0.02$ $[-0.10, +0.21]$. We declare one defect in that amendment: the registered "difficulty" control, the fraction of six public leaderboard runs that solved the instance, turned out to predict our silent failures at 0.95 on its own — one of those runs is our own agent model, and the others fail the same instances — so the registered model was at ceiling and uninformative. The numbers above are from the labelled structure-only sensitivity variant; all three variants are in the record.

4.6 A selector that ranks failures: the regime the proposal named

The study's largest stated limitation was that a 4B chat reward model barely selects. Amendment 5, pre-registered with fixed bars, re-ran the protocol with `gpt-5` scoring *every* candidate in both pools and committing its argmax. No new trajectories; 620 candidates re-scored for $13, and the 59 newly committed patches graded by the same harness.

S-P1 passes on both benchmarks. Across the candidate pool `gpt-5` ranks failing candidates below passing ones at 0.712 [0.62, 0.81] on Verified and 0.706 [0.63, 0.78] on Multilingual, against the 4B model's 0.81 and 0.43. It re-commits 28 of 39 trajectories on Verified and 80 of 106 on Multilingual:

a materially different selected population, chosen by a selector with replicated signal.

S-P2 passes on Multilingual, where the pool score of 0.706 falls to 0.591 on the set it committed — the selection-conditional gap, now measured on a selector that has signal to lose. On Verified it fails: the committed score is 0.792, above the pool, on nine positives and therefore below our inference floor.

S-P3 does not hold. On Multilingual (85 rows, 32 failures, 39 repositories) disagreement reaches AUROC(u) = 0.614, but its interval [0.49, 0.74] includes chance, and the endpoint depends entirely on the estimator: the leave-one-repository-out logistic reports +0.375 [+0.21, +0.52] — and its baseline M_c scores **0.150** out of fold against a raw AUROC($-c$) of 0.591. The apparent gain tracks the baseline's deficit rather than u: baseline 0.150 gives +0.375, baseline 0.377 gives +0.195, and the fixed-weight estimator, whose baseline of 0.588 reproduces the raw value, gives **+0.029** [**−0.08**, **+0.14**]. We report the last. This is the same fragility documented in §4.7, in its most extreme form yet, and it is why that section reports four estimators rather than one.

With a selector that demonstrably ranks failures, that re-commits three-quarters of the population, and whose own signal is measurably consumed by selecting, panel disagreement still adds nothing detectable. The null now holds inside the regime the proposal named.

4.7 The registered endpoint: does u add to c out of fold?

The proposal's confirmatory quantity was the held-out ΔAUROC of a logistic model M_{c+u} over M_c. The 70/30 repository split never held ten test positives, so it was never eligible on Verified alone. With 47 positives across both benchmarks it can be computed by leave-one-repository-out, which we declare as a substitution of the cross-validation scheme, not of the estimand.

A two-parameter model refit on eight to thirty-six clusters is a fragile thing to hang a bound on, so we compute the endpoint four ways (Table 3). Only the first was pre-registered; the other three are post-hoc robustness checks of it, declared as such, and they exist because the first proved fragile: pooled out-of-fold probabilities from a leave-one-repository-out logistic; the same with five clustered folds; AUROC computed *inside* each fold and averaged by fold size, which avoids pooling probabilities across folds with different calibration; and a fixed unit-weight sum of the two standardized scores, which fits no coefficients at all. The four disagree in sign — pooled, they give −0.119, −0.030, −0.014 and +0.013 — and we therefore do not report any one of them as the estimate. What they agree on is the conclusion: every interval includes zero. The bound needs stating carefully. Among the three estimators that use every row, **the loosest pooled upper bound is** +0.12; per benchmark it is +0.25 (Verified, fixed weights), where twelve positives buy little precision. The fourth estimator, which computes AUROC inside each held-out repository, is the weakest: a single repository is usually all-pass or all-fail, so no AUROC is defined there and the fold is dropped — 15 of 46 folds and 53 of 111 rows survive, and its interval, $[-0.31, +0.28]$ pooled, excludes almost nothing. We report it for completeness and do not lean on it. The defensible statement is that adding disagreement to the selector's score produces no detectable gain under any of the four estimators we tested, with a pooled upper bound of +0.12 from those that use all the data.

The spread itself is instructive. The fixed-weight variant reproduces the raw AUROC($-c$) almost exactly (0.559 vs. 0.560 pooled), while the logistic variants report M_c below it (0.48–0.50): pooling out-of-fold probabilities across folds whose intercepts differ scrambles the global ranking, which depresses both arms roughly equally and cancels in the difference. Readers who prefer an assumption-free statement should take the fixed-weight row: +0.013 [−0.07, +0.12] pooled. The discrepancy is worth one sentence beyond this paper: with rank metrics and few clusters, pooling out-of-fold predictions can make a real signal appear to vanish when what has vanished is comparability across refits, and an

Estimator of the registered endpoint	Population	AUROC base	AUROC $+u$	ΔAUROC	95% CI
Logistic, leave-one-repo-out	Verified	0.527	0.451	-0.076	$[-0.16, +0.08]$
(pooled OOF probabilities)	Multilingual	0.459	0.436	-0.022	$[-0.13, +0.08]$
	Pooled	0.484	0.454	-0.030	$[-0.10, +0.04]$
Logistic, 5 clustered folds	Verified	0.644	0.455	-0.189	$[-0.28, +0.07]$
	Multilingual	0.473	0.455	-0.018	$[-0.08, +0.05]$
	Pooled	0.504	0.491	-0.014	$[-0.09, +0.07]$
AUROC within fold, size-weighted	Verified	—	—	-0.152	$[-1.00, +0.40]$
(3/9, 12/37, 15/46 folds usable;	Multilingual	—	—	-0.145	$[-0.38, +0.42]$
22/34, 31/77, 53/111 rows)	Pooled	—	—	-0.119	$[-0.31, +0.28]$
Fixed unit weights, no fitting	Verified	0.614	0.583	-0.030	$[-0.17, +0.25]$
($z(-c) + z(u)$, train-fold scaling)	Multilingual	0.561	0.610	$+0.048$	$[-0.07, +0.18]$
	Pooled	**0.559**	**0.571**	$\mathbf{+0.013}$	$[\mathbf{-0.07}, \mathbf{+0.12}]$

Table 3: The registered confirmatory endpoint under four estimators, repository-clustered bootstrap. Point estimates disagree in sign; every interval includes zero. The within-fold estimator drops repositories where no AUROC is defined and covers about half the rows, so its interval is wide; among the three full-coverage estimators the loosest pooled upper bound is +0.12. Raw AUROC($-c$) is 0.621 / 0.560 / 0.560.

unfitted combination or within-fold evaluation is the cheap check.

4.8 A ruler beats the critics — on one benchmark

Two integers, read off the diff and the transcript with no model at all, rank silent failures better than every reward model at 4k and every disagreement statistic at any length (Fig. 7): lines added in the patch, 0.665; agent step count, 0.657. Bigger patches and longer trajectories fail more. The paired gap over the primary u is +0.16 [−0.11, +0.36], exploratory, on twelve positives. Individual critics do cross this bar once they see the solution at length (§4.3); the panel's disagreement never does on a single window, and adds nothing to the ruler out of fold. We propose lines-added as the baseline any audit statistic must clear before it is worth a confirmatory run — and note that on SWE-bench Verified, beating it by 0.10 in a paired test has power of about 0.3 at any n, because twelve repositories cap the effective sample (§6).

5 Mechanism: stable critics, uninformative disagreement

Draft v0.1 of this paper read Fig. 8 — the same reward model scoring the prefix and the patch of one trajectory with correlation 0.03–0.18 — as evidence that these critics "do not hold a judgment" of code correctness. That reading could not distinguish critic noise from windows that legitimately contain different information, and we pre-registered a test to separate them (Amendment 4b, with the reading fixed in advance). The result withdraws the claim. Re-scoring the identical window after a reformatting that changes no information, every critic reproduces its ranking at $r = 0.98$–0.99; after a 128-token shift, at 0.86–0.95; the per-window panel u itself at 0.91 and 0.73. Adjacent windows with 25% overlap correlate at 0.20–0.56, first against last at −0.39 to 0.47. These critics are stable instruments responding to what is in front of them. The prefix score is a judgment of the problem; the patch score is a judgment of the patch; they differ because the inputs do.

The failure is therefore in the statistic, not the critics. A panel's disagreement is large when its members' rank-normalized scores spread out — which they do for reasons that have nothing to do

with correctness: different critics are calibrated to different aspects of a window, and the spread is dominated by that between-critic heterogeneity rather than by any shared uncertainty about the solution. The signal the critics carry individually (0.71–0.75 at length) is averaged away, not amplified, by measuring how much they disagree. Windowed $u_{\max}$ recovers some of it, plausibly because the maximum over windows behaves partly like a max over the critics' own low scores; its exploratory single-critic analogues (Fig. 5, right) are consistent with that.

The proposal named a mechanism for when a panel should work — errors that are not correlated — and promised to test it by comparing quality-matched panels of differing error correlation. We can now report it. Forming all 35 four-member panels from the seven reward models that scored every candidate at all three lengths (the 32B has no 4k pass), and measuring for each its mean individual AUROC, the mean pairwise correlation of its members' errors, and its AUROC(u), two things emerge. First, and more important: **there is no decorrelated panel to be had.** Every one of the 35 panels has pairwise error correlation between 0.71 and 0.89 on Verified and between 0.74 and 0.84 on Multilingual. The diversity this design assumes one can dial does not exist among off-the-shelf critics — which is what the correlated-judge literature predicts [13, 22, 23], and which we can now state as a measurement on this population rather than an expectation. Second, within that narrow range error correlation does not predict panel performance: the partial correlation with AUROC(u), controlling for panel quality, is -0.11 on Verified and $+0.08$ on Multilingual, opposite in sign and both near zero.

A related check narrows the claim further. If the panel held information that disagreement merely failed to extract, a learned combination of its four raw scores should beat their spread. Out of fold and without fitted weights it does not: 0.43 against 0.50 on Verified, and 0.41 against 0.57 on Multilingual. On the committed population the panel is not carrying something that the wrong statistic discards; it is not carrying much at all.

This explains the earlier false positive too. On HumanEval+ with an LLM-as-judge critic, we briefly observed $u = 0.69$ and a "Goodhart signature" of the critic below chance. Both were artifacts of a critic with three output values on three positives; with a real reward model selecting, both vanished. We withdrew the claim in the record before running the SWE-bench pilot, as we withdraw the instability claim here.

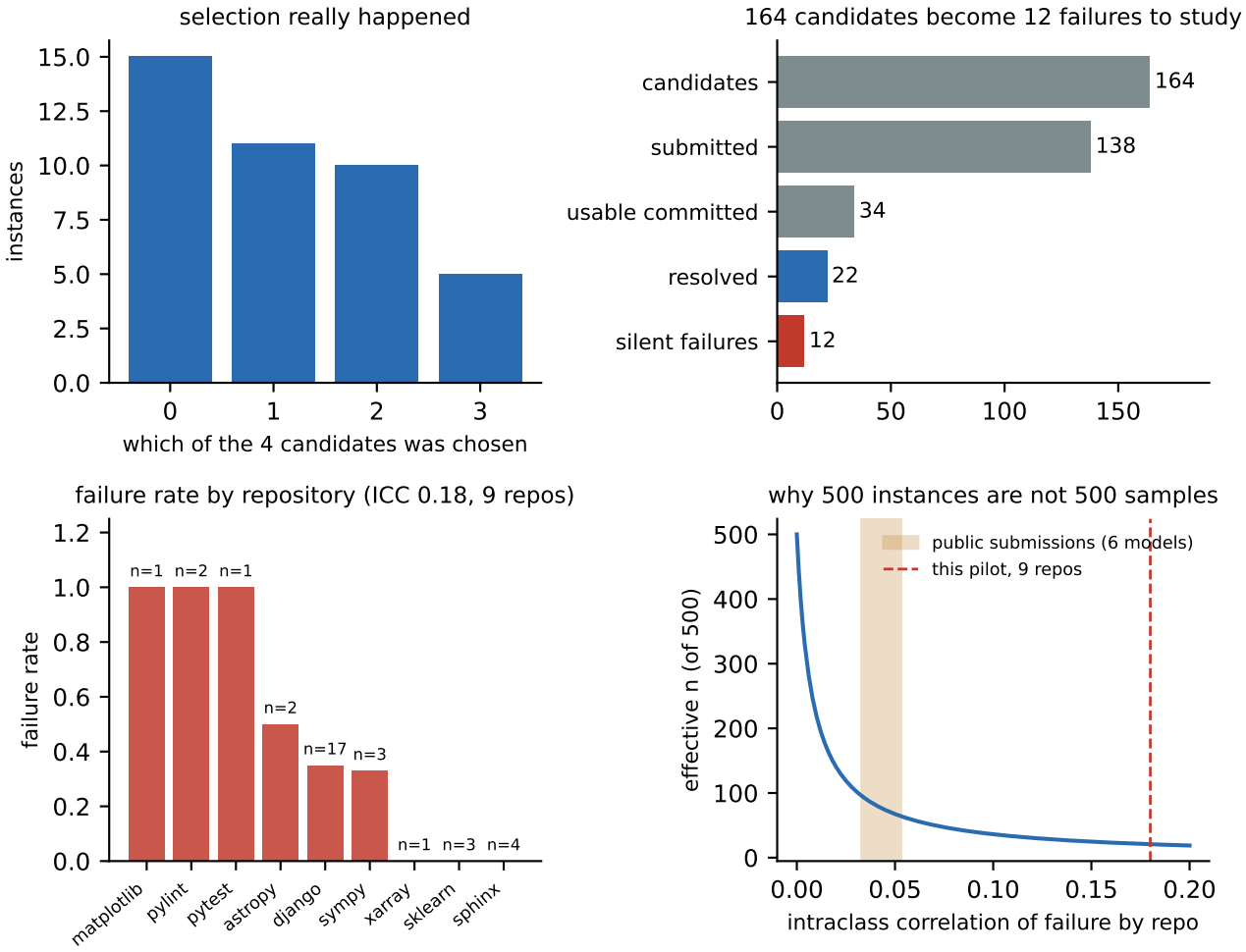


Figure 2: **Where the sample goes, and why it is small.** Top left: best-of-4 with a true reward-model critic genuinely selected (chosen index spread across all four slots; an earlier LLM-judge critic had collapsed to index 0). Top right: from 164 candidates to 12 silent failures. Bottom left: per-repository failure rates, 9 repositories, ICC 0.18. Bottom right: effective sample size on SWE-bench Verified as a function of repository ICC; the public estimate from six leaderboard submissions is 0.033–0.053.

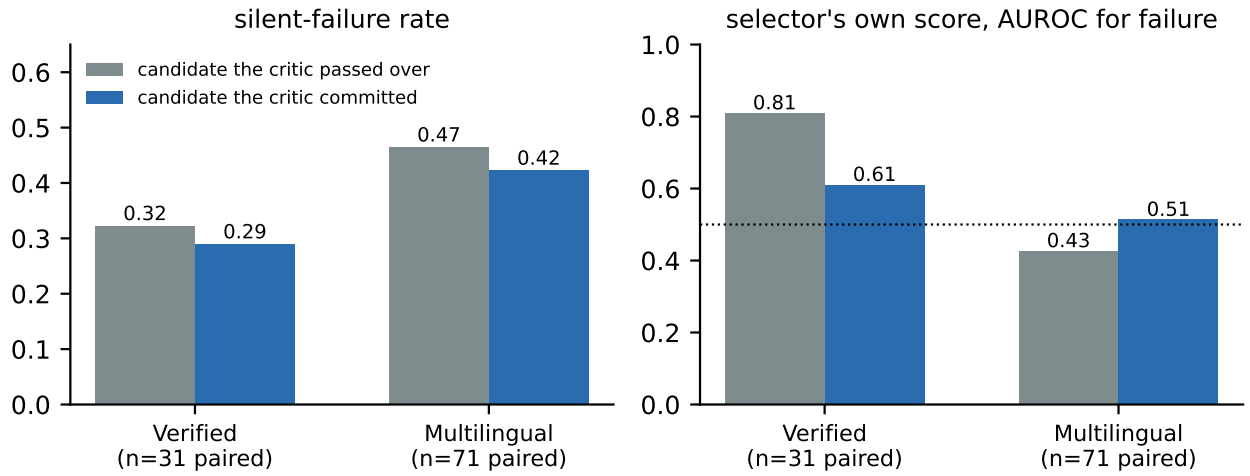


Figure 3: **Picking did not make things worse — but it did use up the picker's own judgment.** One candidate the critic passed over, graded per instance, paired with the one it committed. Left: the committed candidate fails about as often as one passed over. Right: on Verified the selector tells failures from successes among candidates it rejected (0.81) far better than among those it chose (0.61); on Multilingual it could do neither.

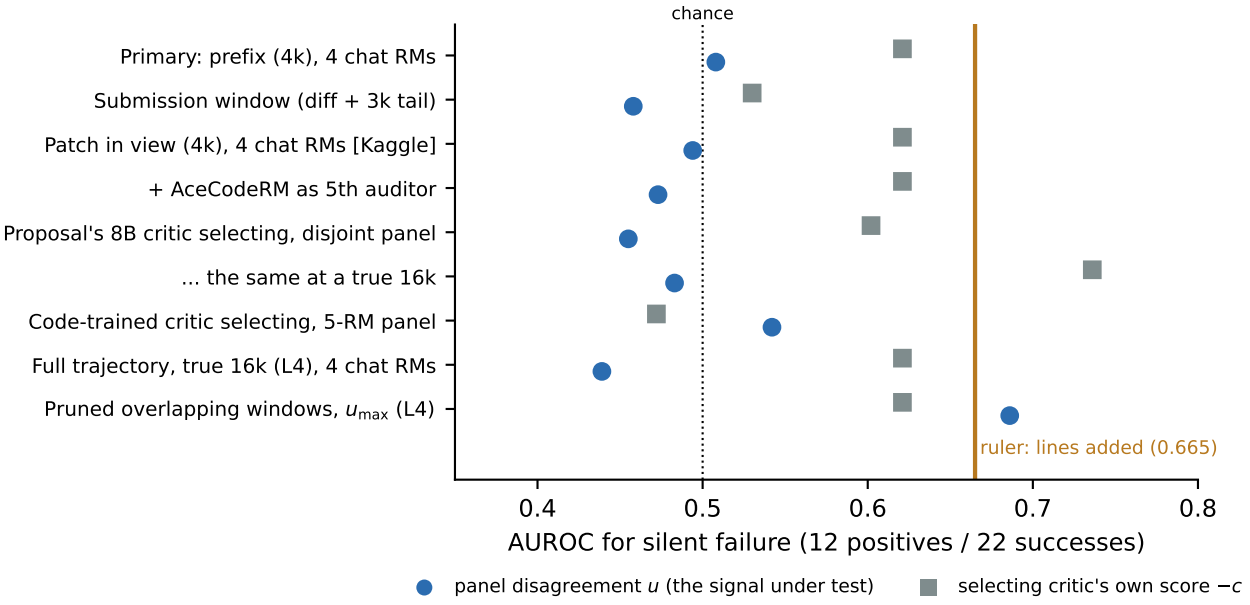


Figure 4: **Every way we tried to make disagreement work, on the pilot.** Each row is one configuration; 0.5 is a coin flip. Circles: panel disagreement u. Squares: the selecting critic's own score. The gold line is the simplest baseline we could find — counting the lines added to the patch. Nothing crosses it here, and the one configuration that later did (Fig. 5) failed to replicate (Fig. 6).

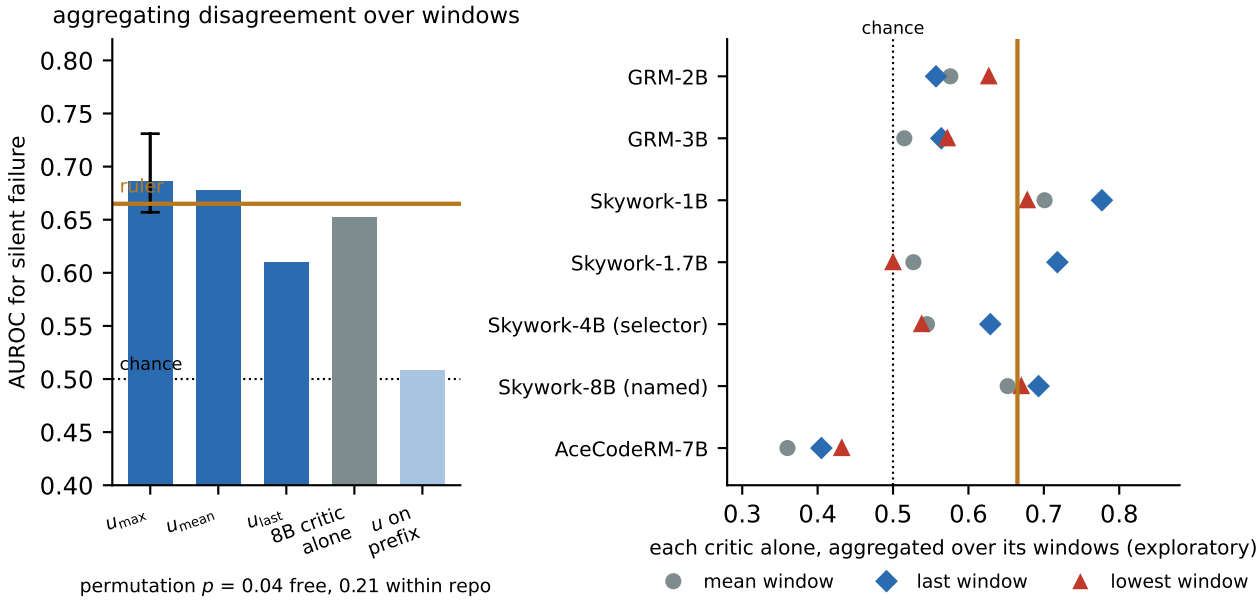


Figure 5: **Letting the panel read the whole trajectory in overlapping chunks.** Left: panel disagreement aggregated over pruned, overlapping windows; whisker on $u_{\max}$ is the leave-one-instance-out range. Right: each critic's own score aggregated over its windows (exploratory; no prediction was registered for these).

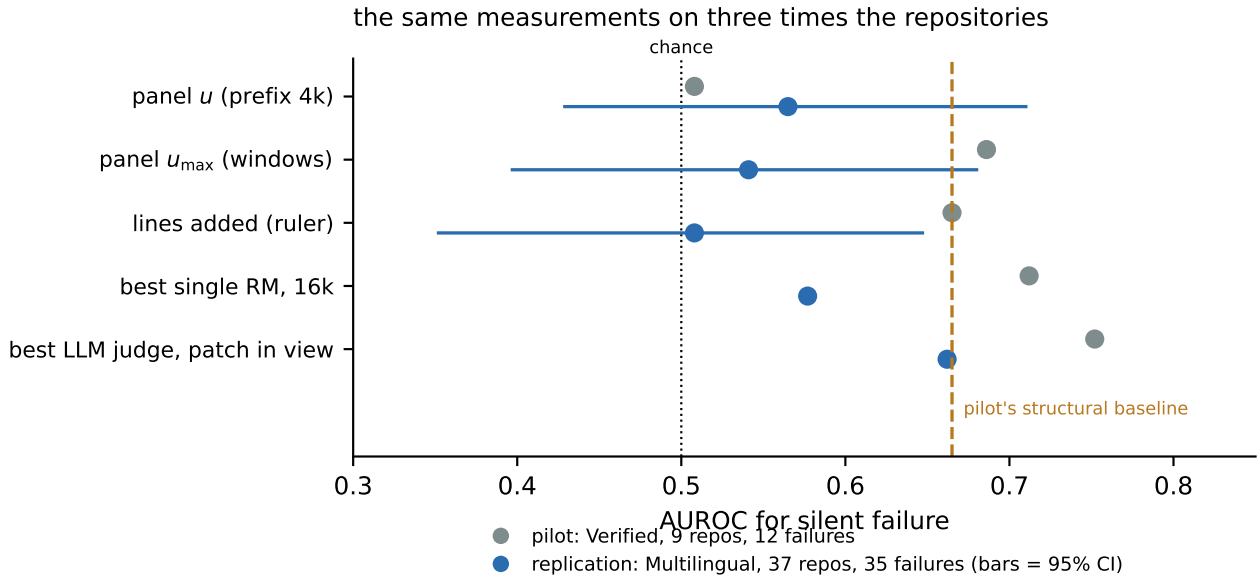


Figure 6: **Everything promising on the small benchmark vanished on the big one.** Grey: the pilot, 9 repositories in the analysed set. Blue: the replication, 37. Every quantity that looked like a signal on Verified, re-measured on SWE-bench Multilingual with the same agent, critic, panel, pruning, windows and executed gold. Intervals are repository-clustered bootstraps.

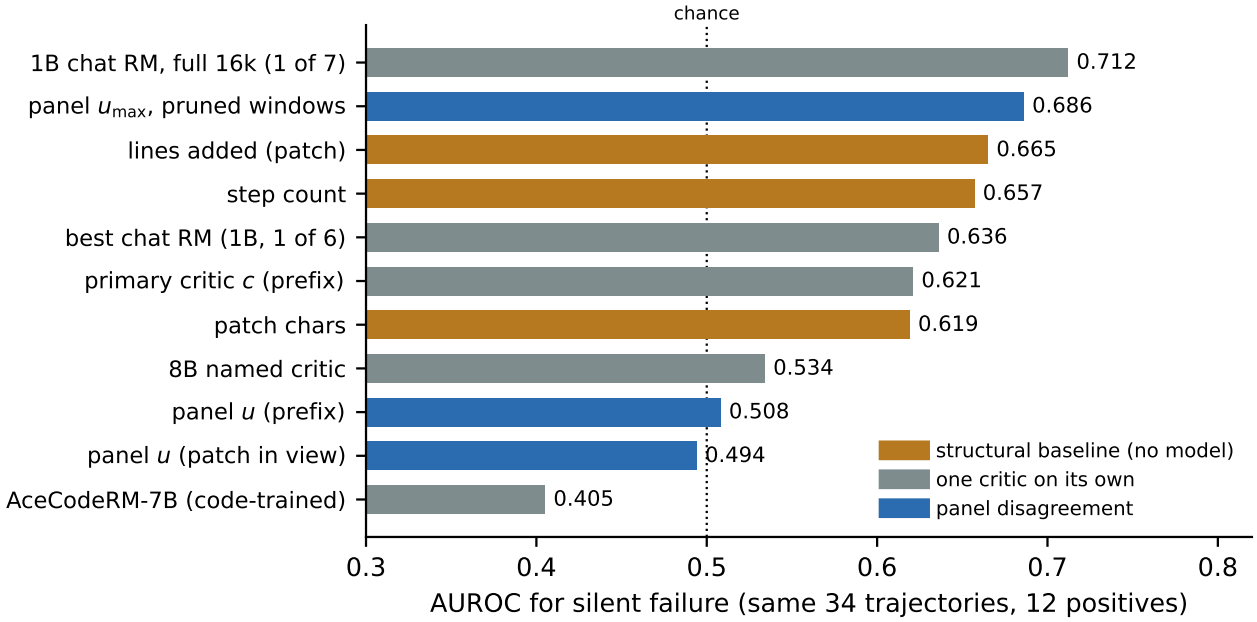


Figure 7: **On the pilot, two integers beat every model we ran.** Same 34 trajectories, same 12 positives. Gold bars need no model at all. Exploratory (declared as such): the covariates were registered as controls, not baselines.

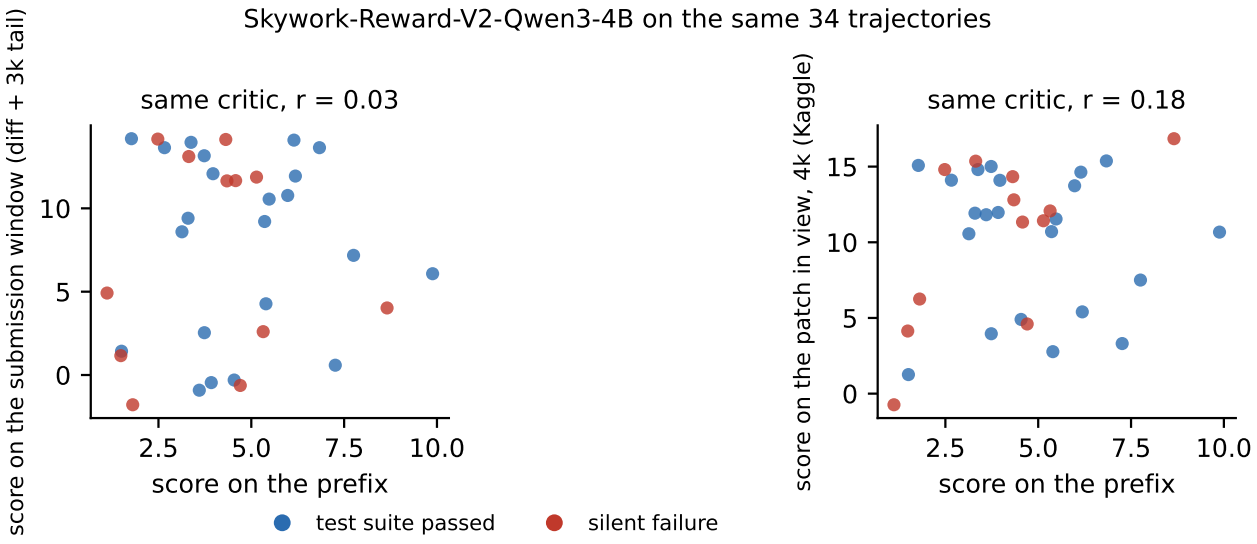


Figure 8: **The same critic, the same trajectory, two different excerpts — and almost unrelated scores.** The primary selecting critic's score on the trajectory prefix (x) against its score on two other windows of the same trajectory (y) — because the windows contain different things (see Fig. 9).

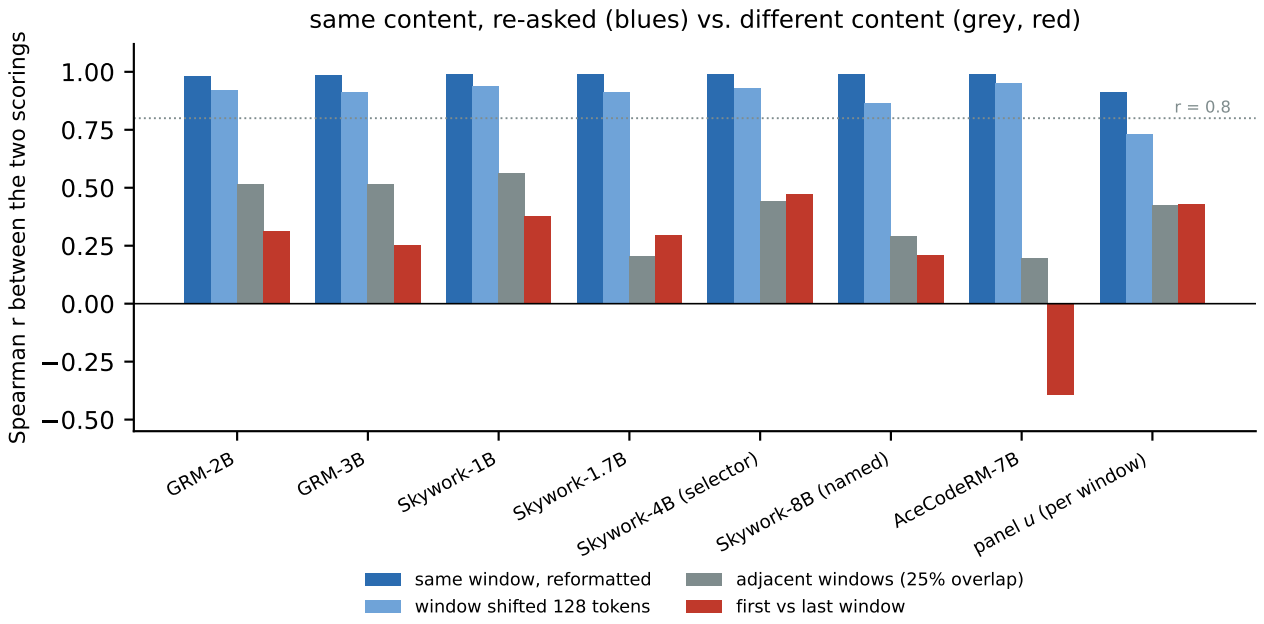


Figure 9: **The critics are not noisy: they react to content.** Re-ask the same critic the same question and it answers the same way; change which part of the trajectory it reads and it does not. Each critic re-scored every Amendment-3 window after a content-preserving reformatting (headers re-cased, pruning marker reworded, one neutral line prepended) and after a 128-token shift. Spearman r over 65–73 windows. The same critic on the same content is essentially deterministic; on adjacent or distant windows it is not.

6 Limitations and declared deviations

- **One agent, one selector family.** gpt-5-mini at 40 steps; a 4B chat reward model selects, so the selected population is that of a weak selector. Failure structure may differ for stronger agents or deployed-grade verifiers.
- **Selection did not concentrate failure here.** The unselected-candidate control (Amendment 4f) shows the committed candidate failing no more often than a passed-over one on either benchmark. The proposal's premise — that selection *enriches* the committed set for critic mistakes — is therefore not demonstrated for this selector; what is demonstrated is that selection removes the selector's own signal (0.81 → 0.61 on Verified). A stronger selector could change both numbers.
- **The selection-conditional regime.** With the primary 4B selector it is observed only on Verified (0.81 unselected to 0.61 committed); on Multilingual that critic had no signal in either arm. Amendment 5 closes this by re-selecting with `gpt-5`, which does rank failures in the pool (0.71 on both benchmarks) and whose signal is measurably consumed by selecting on Multilingual (§4.6). What remains untested is a stronger *agent*: `gpt-5-mini` fails 43.8% of Verified instances against 24–30% for the five stronger public runs on the same benchmark. A capable agent produces fewer failures, so holding 35 positives would need roughly 250 instances rather than 155 — the study gets harder, not easier, and that is the next experiment.
- **Critic classes.** Six chat-preference RMs, two code-trained RMs and three LLM judges from one vendor; the 32B ran at 8k on most rows. Cross-vendor judges and trained trajectory verifiers were not run; the claim is scoped to off-the-shelf critics.
- **Gold is the harness verdict.** It accepts some incorrect patches and rejects some correct ones [40, 42, 53]; failure rates are lower bounds and every AUROC is attenuated by label noise.
- **Forty-seven positives.** Single-AUROC intervals are ±0.25 (Verified) and ±0.14 (Multilingual). Roughly twenty-five analyses were run on the twelve Verified positives without multiplicity correction; the replication is the correction: the configurations that cleared their bars on Verified did not clear them on Multilingual, while the replication's own prediction that the null would hold (R-P1) passed. The study supports failure to reject and an upper bound, not equivalence to chance; on the registered endpoint the bound is +0.04 pooled for the leave-one-repository-out logistic, +0.12 pooled across the three full-coverage estimators of it, and +0.25 on Verified alone (Table 3); on single-protocol comparisons it is +0.22, and the widest interval in the sweep reaches +0.45. We quote the loosest that applies to the population in question.
- **Repository clustering.** Verified is 500 instances over 12 repositories (django 46%); from six public submissions the ICC of resolution by repository is 0.033–0.053, an effective n of 64–95 of 500 (Fig. 2). Clustered bootstraps with nine clusters are anti-conservative; we report null-calibrated coverage in the record. The paired "beat the ruler by 0.10" test has power ≈0.3 on Verified at any n, which is why the replication changed repositories rather than instances.
- **Exploratory analyses.** The diversity-versus-error-correlation comparison, the learned-combiner check, judge calibration, selection pressure at n=2 versus n=4, cross-benchmark transfer, agent self-verification as a feature, and a margin-stratified check were all run after the confirmatory results and are reported as exploratory, without multiplicity correction.
- **Input length and windows.** The primary scored a 4k prefix; the six chat RMs were scored at a true 8k and 16k; `AceCodeRM-7B` fell back to 4k on 153/164 rows. Pruning left most trajectories at one to three windows; $u_{\max}$ is a maximum of noisy statistics and biased upward under the null.
- **Declared deviations.** 41 instances, not 50 (generation stalled on host disk; the missing nine were set by list order). Amendment 4a's registered difficulty control was a near-oracle (0.95) and

was replaced by a labelled structure-only base. The replication's $15 generation cap was exceeded ($18.30; the account's rate limit killed 138 partial trajectories) because stopping at 420/456 would have dropped the harder instances first. 26% of replication candidates submitted no patch; 7 committed patches were ungradable and excluded.

7 Related work

Best-of-n, Goodhart and the winner's curse. Selecting the argmax of a learned proxy over-optimizes it [3, 11]; at inference time the effect appears as reward hacking that grows with n [4, 21, 47] and can be mitigated by pessimism [61]. The statistical form is the optimizer's curse [46]; selecting on a noisy score and then studying the selected set is selection bias in the structural sense [14]. Two recent notes make the evaluation consequence explicit: success counted before an answer is chosen is not deployed success [9], and best-of-n degrades calibration even under a perfect ranker [31]. Our estimand is this consequence for auditors.

Reward-model uncertainty and ensembles. Ensembles mitigate but do not eliminate over-optimization [6, 8]; uncertainty-aware reward models [30] and calibrated process reward models [37] change *selection*. We ask a different question — given the trajectory the selector already committed, does a disjoint panel's disagreement carry residual information — which none of these evaluate.

Correlated judges and panel effective size. Capable models make correlated errors [13, 22]; judge panels supply few effective votes [15, 23]; common-mode error is not identifiable from internal scores [2]; consensus can rival trained verifiers where errors are decorrelated [29, 48] and disagreement can be structured into a confidence estimator [17]. On code, verification helps only on margin-pivotal cases and a single round of debate erases the disagreement it needs [43, 44]. These predict our null; we supply an executed-gold instance on selected agent trajectories.

Process and outcome verifiers. Step-level supervision [26, 52], generative verifiers [64] and agent PRMs [54] score and select; they are not evaluated as residual auditors of a selected set.

SWE agents, verifiers and gold. SWE-bench and its Verified and Multilingual subsets [18, 35, 57], with mini-SWE-agent as the reference scaffold [59], are our testbed; SWE-smith, SWE-Gym and R2E-Gym supply training data and verifiers [16, 36, 58]; Agentless questions the need for agency at all [55]. Execution-free critics and reward models for SWE trajectories [45, 56] are evaluated on discrimination, calibration and test-time scaling over candidate pools; trained failure locators and prefix monitors [39, 51] and sequential-testing verifiers [41] target failure directly. None conditions on an upstream selector. The gold itself is imperfect: accepted patches can fail developer tests, test suites can be weak, and passing runs can be lucky [40, 42, 50, 53].

Agent failure and self-uncertainty. Agents declare success falsely [1], are overconfident about their own success [20], and their early uncertainty does not predict failure [25]; online auditors have been trained for multi-agent settings [63]. Semantic entropy and self-knowledge [10, 19] and selective classification [12] are the uncertainty tools these build on; long-context position effects [28] bear on what a critic sees in a window.

Reward models used. Skywork-Reward-V2 [27], GRM [60] and AceCodeRM [62], chosen by RewardBench standing [24, 32].

Methodology. Paired AUROC comparison for nested models should not use the DeLong test [7, 38]; clustered ROC methods [34] and decision curves [49] inform our analysis. The pre-registration trail follows the registered-report tradition [5, 33].

8 Conclusion

A third to a half of what a reward-model critic confidently committed on two SWE benchmarks was broken. The proposed safeguard — disagreement among a disjoint panel of off-the-shelf critics — did not yield a replicable failure-detection signal under any configuration we pre-registered and ran, across reward models and LLM judges alike, on forty-six repositories. One configuration crossed its bar on the pilot: the maximum disagreement over pruned windows reached 0.686 on twelve positives, with a free permutation p of 0.04 that rose to 0.21 within repository. It did not survive replication. The critics are not noisy: they are stable, content-driven, and given the solution at length a frontier judge ranks silent failures at about two-thirds. What fails is the idea that their *disagreement* is where the missing information lives. The point estimates that suggested otherwise on twelve positives were settled the way they needed to be, on thirty-seven fresh repositories rather than five hundred more trajectories, and they are at chance. Anyone building an escalation trigger for critic-selected agents should evaluate it where it will be used — on the selected population, against the selector's own score, with executed gold — and should expect off-the-shelf disagreement to add nothing detectable: on the registered endpoint, no estimator of it separates from zero, and the loosest bound we can defend on the pooled population is +0.12 AUROC over the selector's own score.

References

[1] Laksh Advani. From confident closing to silent failure: Characterizing false success in llm agents. *arXiv:2606.09863*, 2026. URL https://arxiv.org/abs/2606.09863.

[2] Fariya Afrin and Ibne Farabi Shihab. Evaluator ensembles under reward hacking: Covariance geometry and finite-search guarantees. *arXiv:2608.08002*, 2026. URL https://arxiv.org/abs/2608.08002.

[3] Ahmad Beirami, Alekh Agarwal, Jonathan Berant, Alexander D'Amour, Jacob Eisenstein, Chirag Nagpal, and Ananda Theertha Suresh. Theoretical guarantees on the best-of-n alignment policy. *arXiv:2401.01879*, 2024. URL https://arxiv.org/abs/2401.01879.

[4] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling. *arXiv:2407.21787*, 2024. URL https://arxiv.org/abs/2407.21787.

[5] Christopher D. Chambers and Loukia Tzavella. The past, present and future of registered reports. *Nature Human Behaviour*, 6:29–42, 2022. URL https://doi.org/10.1038/s41562-021-01193-7.

[6] Thomas Coste, Usman Anwar, Robert Kirk, and David Krueger. Reward model ensembles help mitigate overoptimization. *ICLR*, 2024. URL https://arxiv.org/abs/2310.02743. arXiv:2310.02743.

[7] Olga V. Demler, Michael J. Pencina, and Ralph B. D'Agostino. Misuse of delong test to compare aucs for nested models. *Statistics in Medicine*, 31(23):2577–2587, 2012. URL https://doi.org/10.1002/sim.5328.

[8] Jacob Eisenstein, Chirag Nagpal, Alekh Agarwal, Ahmad Beirami, Alex D'Amour, DJ Dvijotham, Adam Fisch, Katherine Heller, Stephen Pfohl, Deepak Ramachandran, Peter Shaw, and Jonathan Berant. Helping or herding? reward model ensembles mitigate but do not eliminate reward hacking. *arXiv:2312.09244*, 2023. URL https://arxiv.org/abs/2312.09244.

[9] Seyed Morteza Emadi. When does a scaling result justify a different allocation? a critical review of resource-allocation evidence for ai systems. *arXiv:2609.14500*, 2026. URL https://arxiv.org/abs/2609.14500.

[10] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024. URL https://doi.org/10.1038/s41586-024-07421-0.

[11] Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. *ICML*, 2023. URL https://arxiv.org/abs/2210.10760. arXiv:2210.10760.

[12] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. *NeurIPS*, 2017. URL https://arxiv.org/abs/1705.08500. arXiv:1705.08500.

[13] Shashwat Goel, Joschka Struber, Ilze Amanda Auzina, et al. Great models think alike and this undermines ai oversight. *arXiv:2502.04313*, 2025. URL https://arxiv.org/abs/2502.04313.

[14] Miguel A. Hernán, Sonia Hernández-Díaz, and James M. Robins. A structural approach to selection bias. *Epidemiology*, 15(5):615–625, 2004. URL https://doi.org/10.1097/01.ede.0000135174.63482.43.

[15] Elias Hossain, Niloofar Yousefi, and Ser-Nam Lim. Agreement overstates evidence: Error dependence in llm judge consensus. *arXiv:2609.22512*, 2026. URL https://arxiv.org/abs/2609.22512.

[16] Naman Jain, Jaskirat Singh, Manish Shetty, Liang Zheng, Koushik Sen, and Ion Stoica. R2e-gym: Procedural environments and hybrid verifiers for scaling open-weights swe agents. *arXiv:2504.07164*, 2025. URL https://arxiv.org/abs/2504.07164.

[17] Bo Jiang. Discouq: Structured disagreement analysis for uncertainty quantification in llm agent ensembles. *arXiv:2603.20975*, 2026. URL https://arxiv.org/abs/2603.20975.

[18] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. Swe-bench: Can language models resolve real-world github issues? *ICLR*, 2024. URL https://arxiv.org/abs/2310.06770. arXiv:2310.06770.

[19] Saurav Kadavath, Tom Conerly, Amanda Askell, et al. Language models (mostly) know what they know. *arXiv:2207.05221*, 2022. URL https://arxiv.org/abs/2207.05221.

[20] Jean Kaddour, Srijan Patel, Gbètondji Dovonon, Leo Richter, Pasquale Minervini, and Matt J. Kusner. Agentic uncertainty reveals agentic overconfidence. *arXiv:2602.06948*, 2026. URL https://arxiv.org/abs/2602.06948.

[21] Hadi Khalaf, Claudio Mayrink Verdun, Alex Oesterling, Himabindu Lakkaraju, and Flavio du Pin Calmon. Inference-time reward hacking in large language models. *arXiv:2506.19248*, 2025. URL https://arxiv.org/abs/2506.19248.

[22] Elliot Kim, Avi Garg, Kenny Peng, and Nikhil Garg. Correlated errors in large language models. *arXiv:2506.07962*, 2025. URL https://arxiv.org/abs/2506.07962.

[23] Guneet Kohli. Nine judges, two effective votes: Correlated errors undermine llm evaluation panels. *arXiv:2605.29800*, 2026. URL https://arxiv.org/abs/2605.29800.

[24] Nathan Lambert, Valentina Pyatkin, Jacob Morrison, et al. Rewardbench: Evaluating reward models for language modeling. *arXiv:2403.13787*, 2024. URL https://arxiv.org/abs/2403.13787.

[25] Zongyue Li, Chengyue Yu, Lei Zang, Chenyi Zhuang, Linjian Mo, and Leilei Gan. Last step matters: Early uncertainty cannot predict failure in long-horizon agents. *arXiv:2608.29685*, 2026. URL https://arxiv.org/abs/2608.29685.

[26] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. *ICLR*, 2024. URL https://arxiv.org/abs/2305.20050. arXiv:2305.20050.

[27] Chris Yuhao Liu, Liang Zeng, Yuzhen Xiao, et al. Skywork-reward-v2: Scaling preference data curation via human-ai synergy. *arXiv:2507.01352*, 2025. URL https://arxiv.org/abs/2507.01352.

[28] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *TACL*, 2024. URL https://arxiv.org/abs/2307.03172. arXiv:2307.03172.

[29] Ning Liu. Llms as a jury: Cross-model consensus can outperform process reward models for llm reasoning. *arXiv:2607.10139*, 2026. URL https://arxiv.org/abs/2607.10139.

[30] Xingzhou Lou, Dong Yan, Wei Shen, Yuzi Yan, Jian Xie, and Junge Zhang. Uncertainty-aware reward model: Teaching reward models to know what is unknown. *arXiv:2410.00847*, 2024. URL https://arxiv.org/abs/2410.00847.

[31] Lauri Lovén, Nam Do, Hassan Mehmood, Dinesh Kumar Sah, and Sasu Tarkoma. The behavioral credibility trilemma: When calibrated autonomy becomes impossible. *arXiv:2605.25739*, 2026. URL https://arxiv.org/abs/2605.25739.

[32] Saumya Malik, Valentina Pyatkin, Sander Land, et al. Rewardbench 2: Advancing reward model evaluation. *arXiv:2506.01937*, 2025. URL https://arxiv.org/abs/2506.01937.

[33] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *PNAS*, 115(11):2600–2606, 2018. URL https://doi.org/10.1073/pnas.1708274114.

[34] Nancy A. Obuchowski. Nonparametric analysis of clustered roc curve data. *Biometrics*, 53(2): 567–578, 1997. URL https://doi.org/10.2307/2533958.

[35] OpenAI. Introducing swe-bench verified. https://openai.com/index/introducing-swe-bench-verified/, 2024.

[36] Jiayi Pan, Xingyao Wang, Graham Neubig, Navdeep Jaitly, Heng Ji, Alane Suhr, and Yizhe Zhang. Training software engineering agents and verifiers with swe-gym. *arXiv:2412.21139*, 2024. URL https://arxiv.org/abs/2412.21139.

[37] Young-Jin Park, Kristjan Greenewald, Kaveh Alim, Hao Wang, and Navid Azizan. Know what you don't know: Uncertainty calibration of process reward models. *arXiv:2506.09338*, 2025. URL https://arxiv.org/abs/2506.09338.

[38] Margaret S. Pepe, Kathleen F. Kerr, Gary Longton, and Zheyu Wang. Testing for improvement in prediction model performance. *Statistics in Medicine*, 32(9):1467–1482, 2013. URL https://doi.org/10.1002/sim.5727.

[39] Salman Rahman, Yubin Kim, Mihir Parmar, et al. Locating hidden failures makes long-horizon agents more reliable. *arXiv:2609.17930*, 2026. URL https://arxiv.org/abs/2609.17930.

[40] Shreshth Rajan. Auditing reward hackability in code rl training environments. *arXiv:2606.16062*, 2026. URL https://arxiv.org/abs/2606.16062.

[41] Shuvom Sadhuka, Drew Prinster, Clara Fannjiang, Gabriele Scalia, Bonnie Berger, Aviv Regev, and Hanchen Wang. E-valuator: Reliable agent verifiers with sequential hypothesis testing. *arXiv:2512.03109*, 2025. URL https://arxiv.org/abs/2512.03109.

[42] Priyam Sahoo, Gaurav Mittal, Xiaomin Li, Shengjie Ma, Benjamin Steenhoek, Pingping Lin, and Yu Hu. Agentlens: Revealing the lucky pass problem in swe-agent evaluation. *arXiv:2605.12925*, 2026. URL https://arxiv.org/abs/2605.12925.

[43] Yang Shu. Blind to the pivotal vote: Aggregate independence metrics miss where verification actually helps. *arXiv:2608.06940*, 2026. URL https://arxiv.org/abs/2608.06940.

[44] Yang Shu. Unanimity without persuasion: A single round of debate erases the disagreement that verification needs. *arXiv:2609.26145*, 2026. URL https://arxiv.org/abs/2609.26145.

[45] KaShun Shum, Binyuan Hui, Jiawei Chen, Lei Zhang, Jiaxi Yang, Yuzhen Huang, Junyang Lin, Junxian He, et al. Swe-rm: Execution-free feedback for software engineering agents. *arXiv:2512.21919*, 2025. URL https://arxiv.org/abs/2512.21919.

[46] James E. Smith and Robert L. Winkler. The optimizer's curse: Skepticism and postdecision surprise in decision analysis. *Management Science*, 52(3):311–322, 2006. URL https://doi.org/10.1287/mnsc.1050.0451.

[47] Benedikt Stroebl, Sayash Kapoor, and Arvind Narayanan. Inference scaling flaws: The limits of llm resampling with imperfect verifiers. *arXiv:2411.17501*, 2024. URL https://arxiv.org/abs/2411.17501.

[48] Pat Verga, Sebastian Mazzola, Ruhui Lyu, et al. Replacing judges with juries: Evaluating llm generations with a panel of diverse models. *arXiv:2404.18796*, 2024. URL https://arxiv.org/abs/2404.18796.

[49] Andrew J. Vickers and Elena B. Elkin. Decision curve analysis: A novel method for evaluating prediction models. *Medical Decision Making*, 26(6):565–574, 2006. URL https://doi.org/10.1177/0272989X06295361.

[50] Binghai Wang, Chenlong Zhang, Dayiheng Liu, et al. The verification horizon: No silver bullet for coding agent rewards. *arXiv:2606.26300*, 2026. URL https://arxiv.org/abs/2606.26300.

[51] Chenyu Wang, Yunbo Lyu, Junda He, Zhou Yang, Chenxing Zhong, Yaniv Harel, and David Lo. Fail-fast, restart-smart: Early failure prediction and restart for swe agentic tasks. *arXiv:2608.03222*, 2026. URL https://arxiv.org/abs/2608.03222.

[52] Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. Math-shepherd: Verify and reinforce llms step-by-step without human annotations. *ACL*, 2024. URL https://arxiv.org/abs/2312.08935. arXiv:2312.08935.

[53] You Wang, Michael Pradel, and Zhongxin Liu. Are "solved issues" in swe-bench really solved correctly? an empirical study. *arXiv:2503.15223*, 2025. URL https://arxiv.org/abs/2503.15223.

[54] Zhiheng Xi, Chenyang Liao, Guanyu Li, et al. Agentprm: Process reward models for llm agents via step-wise promise and progress. *arXiv:2511.08325*, 2025. URL https://arxiv.org/abs/2511.08325.

[55] Chunqiu Steven Xia, Yinlin Deng, Soren Dunn, and Lingming Zhang. Agentless: Demystifying llm-based software engineering agents. *arXiv:2407.01489*, 2024. URL https://arxiv.org/abs/2407.01489.

[56] Aashish Yadavally, Hoan Nguyen, Laurent Callot, and Gauthier Guinet. Large language model critics for execution-free evaluation of code changes. *arXiv:2501.16655*, 2025. URL https://arxiv.org/abs/2501.16655.

[57] John Yang, Carlos E. Jimenez, et al. Swe-bench multilingual. *SWE-bench project, https://www.swebench.com/multilingual.html*, 2025. URL https://www.swebench.com/multilingual.html.

[58] John Yang, Kilian Leret, Carlos E. Jimenez, Alexander Wettig, Kabir Khandpur, Yanzhe Zhang, Binyuan Hui, Ofir Press, Ludwig Schmidt, and Diyi Yang. Swe-smith: Scaling data for software engineering agents. *arXiv:2504.21798*, 2025. URL https://arxiv.org/abs/2504.21798.

[59] John Yang, Kilian Lieret, et al. mini-swe-agent. *https://github.com/SWE-agent/mini-swe-agent, version 2.4.6*, 2025. URL https://github.com/SWE-agent/mini-swe-agent.

[60] Rui Yang, Ruomeng Ding, Yong Lin, Huan Zhang, and Tong Zhang. Regularizing hidden states enables learning generalizable reward model for llms. *NeurIPS*, 2024. URL https://arxiv.org/abs/2406.10216. arXiv:2406.10216.

[61] Zhuohao Yu, Zhiwei Steven Wu, and Adam Block. From curiosity to caution: Mitigating reward hacking for best-of-n with pessimism. In *ICLR*, 2026. URL https://arxiv.org/abs/2604.04648. arXiv:2604.04648.

[62] Huaye Zeng, Dongfu Jiang, Haozhe Wang, Ping Nie, Xiaotong Chen, and Wenhu Chen. Acecoder: Acing coder rl via automated test-case synthesis. *arXiv:2502.01718*, 2025. URL https://arxiv.org/abs/2502.01718.

[63] Boxuan Zhang, Jianing Zhu, Zeru Shi, Dongfang Liu, and Ruixiang Tang. Agentforesight: Online auditing for early failure prediction in multi-agent systems. *arXiv:2605.08715*, 2026. URL https://arxiv.org/abs/2605.08715.

[64] Lunjun Zhang, Arian Hosseini, Hritik Bansal, Mehran Kazemi, Aviral Kumar, and Rishabh Agarwal. Generative verifiers: Reward modeling as next-token prediction. *arXiv:2408.15240*, 2024. URL https://arxiv.org/abs/2408.15240.

A Chronology and cost

Stage	What	Cost
Freezes, defects, calibrated power	18 regression tests	$0
HumanEval+ pilots 1–3b (instrument validation)	40 tasks, $n \in \{4, 16\}$, 2 judges → 4 RMs → RM selection	$0
SWE-bench infrastructure without admin rights	Colima + Rosetta + swebench 5.x; three oracle controls	$0
SWE-bench pilot	164 gpt-5-mini candidates, 41 instances, official gold	$3.65
Amendment 1 (submission window)	local, 5 RMs	$0
Amendment 2 (Kaggle 2×T4)	7 critics, 164 candidates, two re-selections graded	$0
Amendments 2 (true 16k) and 3 (Algoverse L4)	7 critics × 164 candidates; 7 critics × 34 windowed rows	$0
Amendments 4a, 4b, 4d (local; L4 stage 5)	incremental-value model; test–retest 7 critics × 3 variants; 8k ladder point	$0
Amendment 4c (OpenAI API)	3 judges × 2 conditions × 34	$1.96
Modal (A100-80GB; CPU sandboxes)	AceCodeRM-32B × 164 + windows; official gold for 9 re-selected patches	≈$3
Amendment 4f control (Modal)	official gold for 144 unselected candidates, both benchmarks	≈$8
Amendment 4e replication (OpenAI + Modal)	456 trajectories, 114 instances, 41 repos; 7 critics at 4k/16k/windows; gold with 3 oracle pairs	$18.30 + ≈$8

B Glossary and the shape of the study

Reading the figures. Colour means the same thing throughout: **blue** is panel disagreement, the signal under test; **grey** is a single critic acting alone; **gold** is a baseline that needs no model at all (lines added to the patch); **red** marks silent failures or the most distant comparison in a panel. Every AUROC axis carries a dotted line at 0.5, marked *chance*; a gold line, where one appears, is the structural baseline a signal has to beat to be worth deploying.

Term	Meaning in this paper
trajectory	one complete attempt by the agent at one task: everything it read, ran and wrote, ending in a patch
critic (reward model, judge)	a model that reads a trajectory and returns a number meaning "how good is this". Off-the-shelf; not trained by us
best-of-n selection	generate n attempts, score each with one critic, keep the highest. Here $n = 4$
committed / silent failure	the attempt the critic kept; a silent failure is one the agent declared finished, the critic ranked best, and the hidden tests reject
c	the selecting critic's score of the committed trajectory
u (disagreement)	how much the audit panel disagrees about the committed trajectory: the spread of members' scores after each is put on a common scale
gold	the verdict of the project's own test suite, run in a container. Not a model's opinion
AUROC	the chance a randomly chosen failure is ranked above a randomly chosen success. 0.5 is a coin flip, 1.0 is perfect
pre-registration	writing down the analysis and its pass/fail thresholds, dated, before the data exists

Stage	What we expected	What happened
Proposal	disagreement would flag failures the selector missed	—
Pilot (Verified)	—	35% silent failures; disagreement at chance; patch size beat every critic
Amendments 1–2	a longer look, or a better critic, would help	neither did, across 4k–16k and eight critics
Amendment 3	reading the trajectory in windows would help	the only configuration that ever crossed the bar, on 12 failures
Amendments 4a–4d	structure and stability diagnostics	no critic adds value beyond patch size; critics are stable, so the earlier "instability" explanation was wrong and was withdrawn
Amendment 4e	replicate on 41 repositories (37 analysed)	every promising Verified number fell to chance, including the windows and patch size
Amendment 4f	selection concentrates failures	it does not; but it does consume the selector's own signal
Registered endpoint	—	disagreement adds nothing to the selector: pooled bound +0.12 AUROC

C Operating points and alternative disagreement statistics

Exploratory; not pre-registered. The escalation-trigger framing implies a review budget. Table 4 gives recall and precision when the top 5/10/20% of committed trajectories by each signal are flagged (positives: 12 of 34; 35 of 77). Table 5 replaces the registered u (sample SD of four rank-normalized scores) with other functions of the same panel scores; the highest is 0.601.

		recall at budget			precision at budget		
Benchmark	Signal	5%	10%	20%	5%	10%	20%
Verified	panel u	0.08	0.08	0.25	0.50	0.33	0.43
	$-c$	0.17	0.17	0.33	1.00	0.67	0.57
	lines added	0.08	0.17	0.42	0.50	0.67	0.71
	gpt-5 judge	0.08	0.17	0.33	0.50	0.67	0.57
Multilingual	panel u	0.06	0.06	0.23	0.50	0.25	0.53
	$-c$	0.11	0.17	0.29	1.00	0.75	0.67
	lines added	0.03	0.09	0.17	0.25	0.38	0.40
	gpt-5 judge	0.09	0.14	0.23	0.75	0.63	0.53

Table 4: Review-budget operating points. Prevalence 0.35 and 0.45; a random flag has precision equal to prevalence.

	SD (registered)	range	mean abs. dev.	entropy	−mean rank	−min rank
Verified	0.508	0.559	0.481	0.500	0.466	0.483
Multilingual	0.565	0.587	0.552	0.393	0.601	0.581

Table 5: AUROC for silent failure of alternative statistics of the same four rank-normalized panel scores (exploratory).

D Reproducibility

Code, pre-registrations with amendments, every scored row, every critic's scores, the harness reports and the research log are in the repository (`UPDATES.md` has one entry per stage: done / learned / changed). All figures are generated by `paper/make_figures.py` from committed JSON.